EXP 4

Strategize impedance matching with characteristic impedance and load impedance through the measurement of a) Location of the stub b) Length of the stub

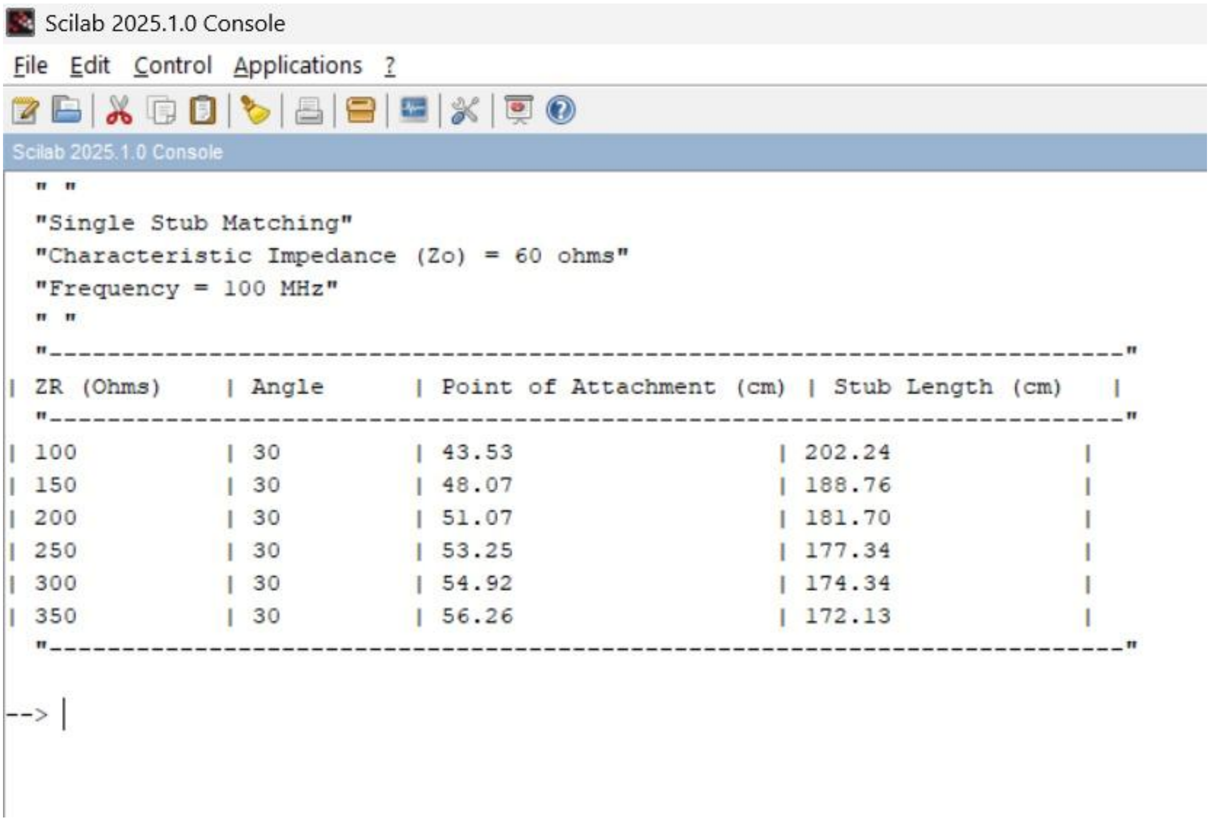
TEST CASE 1 :

PROGRAM

```
clear;
clc;
// Constants
Zo = 60;          // Characteristic impedance (ohms)
f = 100*10^6;     // Frequency (Hz)
theta = 30;       // Load angle (degrees)
lambda = 300/(f*10^-6);   // Wavelength (meters)
// Load impedance values
ZR = [100 150 200 250 300 350];
disp(" ");
disp("Single Stub Matching");
disp("Characteristic Impedance (Zo) = 60 ohms");
disp("Frequency = 100 MHz");
disp(" ");
disp("------------------------------------------------------------------------");
mprintf("| %-12s | %-10s | %-22s | %-18s |\n",...
       "ZR (Ohms)","Angle","Point of Attachment (cm)","Stub Length (cm)");
disp("------------------------------------------------------------------------");
for i = 1:length(ZR)
   Ls = (lambda/(2*%pi))*atan(sqrt(ZR(i)/Zo));
   Lt = (lambda/(2*%pi))*(%pi + atan((sqrt(ZR(i)*Zo))/(ZR(i)-Zo)));
```

```
    mprintf("| %-12d | %-10d | %-22.2f | %-18.2f |\n",...
            ZR(i),theta,Ls*100,Lt*100);
end
disp("------------------------------------------------------------------------");
```

OUTPUT :



```
Scilab 2025.1.0 Console
File  Edit  Control  Applications  ?
Scilab 2025.1.0 Console
  " "
  "Single Stub Matching"
  "Characteristic Impedance (Zo) = 60 ohms"
  "Frequency = 100 MHz"
  " "
  "------------------------------------------------------------------------"
| ZR (Ohms)     | Angle        | Point of Attachment (cm)  | Stub Length (cm)    |
  "------------------------------------------------------------------------"
| 100          | 30         | 43.53                  | 202.24             |
| 150          | 30         | 48.07                  | 188.76             |
| 200          | 30         | 51.07                  | 181.70             |
| 250          | 30         | 53.25                  | 177.34             |
| 300          | 30         | 54.92                  | 174.34             |
| 350          | 30         | 56.26                  | 172.13             |
  "------------------------------------------------------------------------"

--> |
```

TEST CASE 2

PROGRAM:

```
clear;
clc;
// Constants
ZR = 100;              // Load impedance (ohms)
f = 100 * 10^6;        // Frequency (Hz)
theta = 30;            // Load angle (degrees)
lo = 300/(f*(10^-6));  // Wavelength (meters)
// Characteristic impedance values
Zo = [600 800 1000 1200 1400 1600 1800];
disp(" ");
disp("Single Stub Matching");
disp("Load Impedance (ZR) = 100∠30° ohms");
disp("Frequency = 100 MHz");
disp(" ");
disp("----------------------------------------------------------------------------");
mprintf("| %-10s | %-25s | %-25s |\n",...
       "Zo (ohms)","Point of Attachment (cm)","Stub Length (cm)");
disp("----------------------------------------------------------------------------");
for i = 1:length(Zo)
   Ls = (lo/(2*%pi)) * atan(sqrt(ZR/Zo(i)));
   Lt = (lo/(2*%pi)) * (%pi + atan((sqrt(ZR*Zo(i)))/(ZR-Zo(i))));
   mprintf("| %-10d | %-25.2f | %-25.2f |\n",...
          Zo(i), Ls*100, Lt*100);
end

disp("----------------------------------------------------------------------------");
```

OUTPUT :

```
Scilab 2025.1.0 Console
File  Edit  Control  Applications  ?
Scilab 2025.1.0 Console
  " "
  "Single Stub Matching"
  "Load Impedance (ZR) = 100∠30° ohms"
  "Frequency = 100 MHz"
  " "
  "--------------------------------------------------------------------------------"
| Zo (ohms)    | Point of Attachment (cm)  | Stub Length (cm)            |
  "--------------------------------------------------------------------------------"
| 600          | 18.51                     | 128.25                      |
| 800          | 16.23                     | 131.67                      |
| 1000         | 14.62                     | 133.87                      |
| 1200         | 13.42                     | 135.43                      |
| 1400         | 12.47                     | 136.62                      |
| 1600         | 11.70                     | 137.56                      |
| 1800         | 11.05                     | 138.32                      |
  "--------------------------------------------------------------------------------"

-->
```